

TLA+ Study Group — Week 01

Bartosz

November 22, 2025

Brief Description of the Topics

Lecture 1

What is TLA+, for what can it be used and how it's helpful.

- A tool for specification, finding and correcting design errors.
- Well suited for parallel and distributed programming.
- TLA+ sits at the design level/above the code level.
- It abstracts less-critical/lower level parts of systems.
- The properties TLA+ can check are conditions on individual executions (i.e. 99% of executions are correct is not a property testable by TLA+)
- Execution of a system is represented as a sequence of discrete steps
- TLA+ represents systems as state machines.

Lecture 2

State Machines and TLA+ notation.

- TLA+ operates on formulas not statements.

Language keywords

TRUE true

FALSE false

$\wedge$ logical and

$\vee$ logical or

= equality

== $\triangleq$

' next state i.e. $pc' = pc + 1$

EXTENDS module import

VARIABLES variable declarations

Lecture 3

TLA+ resources.

Some links to TLA+ groups and introduction materials but mostly a HOWTO on how to install TLA+ and run a model.

All that was missing was techno music and instructions written live in notepad.

What stood out

0.1 Lecture 1

... not solutions but recognizing **important** problems that had been obscured by irrelevant details

- The process of simplicifaction by removing irrelevant details is called abstraction.
- *TLA+ has added significant value, either finding subtle bugs we would not have found by other means, or giving us enough confidence to make aggressive optimizations without sacrificing correctness*
- *We witnessed first hand the brain washing done by years of C programming – amen*
- Parallel and distributed systems being represented by sequential processes is kinda obvious (Parallel Turing Machines are exactly as powerful as Turing Machines) yet profound.

0.2 Lecture 2

- Mathematical notation instead of PL notation.
- Variables and identifiers must be declared before they are used – suggesting a single pass language design

0.3 Lecture 3

Open Questions